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Developing a child-reported measure of inpatient experience of healthcare.

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Abstract

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There is a gap between children's right to report on their own experience of inpatient care, and availability of a validated measure to facilitate consistent reporting by children and young people. This study aimed to understand children and young peoples (aged 12-18 years) preferences for reporting their own experience of inpatient health care, and validate a revised question set. A three-phase study assessed: content analysis of focus group discussions; face and content validity of an adapted question set; construct validity using confirmatory factor analysis. Children and young people chose to adapt a survey titled 'Australian Hospital Patient Experience Question Set'. Suggested changes to wording reflected their experiences of feeling listened to, heard, and safe, in contrast to feeling overlooked or overpowered. Assessment of construct validity demonstrated a sound one-factor model (n=193, Root Mean Square Error of Approximation 0.033, Comparative Fit Index 0.997, Tucker Lewis Index 0.996, composite reliability 0.951). Children and young people valued participating in survey design. The resulting question set is a reliable and valid tool to measure self-reported experience of care for children and young people aged 12 to 17 years on their own or with their parent.

Keywords

Quality of Care; Questionnaire design; Benchmarking, Health care; Pediatric healthcare, Patient reported experience measure.

Introduction

United Nations Convention on the Rights of the Child (UNCRC) emphasises a fundamental right of children and young people to express their views, to be heard, and to be taken seriously (United Nations, 1989). A key aspect of this is that children and young people (CYP), as health care consumers, can report on their experience of health care from their perspective (Davies et al., 2023). However, CYPs views of their own health care experience are rarely measured; rather parents are asked to report for, or with, CYP (Starlight Children's Foundation, 2021). A gap between CYPs right to report on their own experience of inpatient care and availability of a validated measure to facilitate consistent reporting by CYP is internationally reported (Ali et al., 2020; Bele et al., 2021).

Feedback on experience of care is used to improve quality of care; to measure experience of actual care that was received, reflecting how well organisational structures and processes support caregiving, and patient and family engagement with care (Larson et al., 2019). This differs to satisfaction with care. Satisfaction is a subjective measure, reflecting patient and family expectations but not accounting for structures and processes that support delivery of care. World Health Organization (WHO) (2018: 7) theoretical "framework for improving the quality of paediatric care" (WHO Framework) has eight domains organised in three main themes. Three domains within theme 'experience of care' are "effective communication and meaningful participation; respect, protection, and fulfilment of child rights; emotional and psychological support". Consistent with experience of care, this model

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demonstrates interpersonal aspects of care within a framework of available human and physical resources, evidence-based practices, information systems, and referral systems (World Health Organization, 2018).

The WHO Framework recognises that health, developmental, psychosocial, communication, and physical needs of CYP are different to those of adults (World Health Organization, 2018). Demonstrated by Hargreaves et al. (2018), Children and Young People's survey (© Picker Institute Europe) was distributed to parents and children aged eight to 15 years in one set of questions (Hargreaves et al., 2018). In comparison to report by parent-proxy, CYP reported lower involvement in their care, poorer experience of communication with staff, and higher experience of pain (Hargreaves et al., 2018). This emphasises value in direct feedback of health care experience by CYP, to ensure integrity and accuracy of measurement and improve quality of care (World Health Organization, 2018). Parents and carers often make decisions on behalf of their children (Bagattini, 2019). However, this must be balanced with CYPs right to express their views, to be heard, to be taken seriously, and to participate in health care (Davies et al., 2023).

In recognition of different contextual factors of paediatric inpatient care, Australian Hospital Patient Experience Question Set for parents (AHPEQS-Parent) was developed and validated with and for parents of inpatient children (Nelson et al., 2023). AHPEQS-Parent was adapted from AHPEQS, an experience of care survey structured around a conceptual framework identified through focus groups with adult consumers (Australian Commission on Safety and Quality in Health Care, 2017). In qualitative analysis to adapt AHPEQS-Parent, three domains of parent experience were identified: effective communication, respect, and emotional support, domains aligned with the WHO Framework (Nelson et al., 2023). Consistent with Hargreaves et al. (2018) emotional support included a focus on assessment for pain or distress (World Health Organization, 2018).

In focus group discussions (FGDs) parents reported an importance for CYP to have their own experience of care survey (Nelson et al., 2023). This is consistent with an internationally reported gap in non-disease-specific, validated child reported measures of inpatient experience (Ali et al., 2020; Bele et al., 2021). This article reports a multi-methods study to develop a CYP-reported measure of inpatient experience of health care.

Aim

To understand children and young peoples (aged 12 to 18 years) preferences for reporting their own experience of inpatient health care and validate a resulting question set.

METHODS

Design

Research was guided by WHO "framework for improving the quality of paediatric care" (World Health Organization, 2018).

In this study children and young people assessed their own preferences for reporting health care experience. "Child" refers to from birth to 15 years of age and "young person" refers to from 16 to 18 years of age (World Health Organization, 2018). We

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did not assume that a revised version of AHPEQS would meet CYPs preference for reporting their own experience of health care. For this reason, our study began with a qualitative focus to hear CYP preference.

This three-phase mixed-methods design included focus group discussions (FGDs) with CYP, and content analysis of focus group data. Second, a preferred question set was assessed for face validity by CYP and content validity by service providers. Third, construct validity and reliability (Figure 1). Ethics approval was obtained from health service Human Research Ethics Committee (RGS 04554). Standards for Quality Improvement Reporting Excellence were used (Ogrinc et al., 2016).

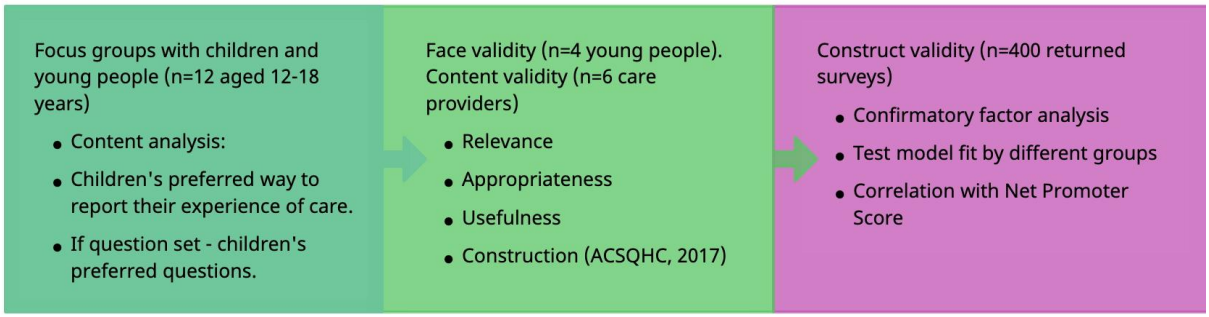


Figure 1. Content of the three-phase research process

Study setting

Western Australia, the largest state of Australia, has a land area of 975,685 square miles. Its 298-bed tertiary hospital for CYP, aged from neonate to 18 years, located in Perth capital city, serves CYP and families from remote, rural, and metropolitan regions. Families include first nations Aboriginal families, and families new to Australia who speak a language other than English at home.

Measures

AHPEQS. A 12-question patient experience survey, developed with adult consumers and validated using rigorous methodology for use in adult inpatient setting (Australian Commission on Safety and Quality in Health Care, 2017).

An aligned AHPEQS Easy English version includes visual prompts and simplified language to facilitate consistent measurement of health care experience. Most questions are answered on a Likert scale, scored from 1 (‘always’) to 5 (‘never’), for example, Question 2, ‘My individual needs were met.’ These factual response options are most appropriate to facilitate objective experience of care measurement (Larson et al., 2019). Two questions use branching logic. For example, ‘Staff made sure you got everything you needed.’ An answer of ‘sometimes’ ‘rarely’ or ‘never’ leads to question ‘Staff told you why they could not give you what you needed.’ We worked with parents to adapt AHPEQS and identify additional questions specific to parent experience (Nelson et al., 2023). AHPEQS-Parent demonstrated sound construct validity, representing a one-factor model of experience (Composite reliability=0.962) (Nelson et al., 2023).

Net Promoter Score (NPS). NPS (Reichheld, 2003) is routinely distributed with AHPEQS and AHPEQS-Parent in Western Australia, asking one question ‘How likely

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are you to recommend (name of health service) to family or friends?' NPS is answered on a scale of 0 ('not at all likely') to 10 ('very likely'). Correlation of NPS and an overall quality of care rating was assessed independently of the sum of AHPEQS-Child.

Content Validity Index (CVI). In phase 2, identified questions were scored 1 (*not*) to 4 (*high*) against a set of quality criteria using CVI (Lynn, 1986). A rating of 3 or 4 by six experts at 83% represented content validity (Lynn, 1986). Questions that did not meet a recommended minimum of 83% for one or more of 16 quality criteria were amended until experts identified an acceptable CVI.

Analysis

Analysis for each phase is reported, beginning with study Phase 1.

Phase 1: Content analysis of focus group data.

In Phase 1, CYP participated in semi-structured focus group discussions. Purposive sampling included CYP who had previous inpatient experience and who spoke English language. Participants were invited via health service social media platforms and consumer engagement network. A participant information sheet was sent by email, written consent was obtained. In line with health service Consumer Representative Recruitment and Management Policy, each participant received a gift voucher in recognition of their expertise and time. Each focus group was open to CYP aged from 12 to 18 years, reflecting existing practice of the health service youth advisory group. To minimise bias, FGDs were moderated by a research team member who had extensive experience in facilitating consumer consultation and who was not included in development of AHPEQS-Parent (HH). A second facilitator was present (HN). Data were recorded, transcribed verbatim, deidentified, and stored securely according to Health Service and ethical guidelines.

A focus group discussion guide included potential questions: 1) "Can you tell us things that are most important when you have stayed in hospital?" 2) An invitation to comment on suitability of AHPEQS and AHPEQS Easy English for ages 12-18 years, was derived from previous research (Australian Commission on Safety and Quality in Health Care, 2019). 3) How CYP would prefer to talk about their experience of care, including how to distribute a survey. 4) Was there anything else we should know about asking children about their stay in hospital?

Content analysis. Qualitative content analysis was conducted using qualitative methods with a deductive approach, categorizing data into domains based on themes (Bengtsson, 2016). Manifest analysis involved analysing words spoken by CYP (Bengtsson, 2016). Steps of analysis: 1. Coding Framework: Utilized three domains of 'experience of healthcare' within WHO Framework to describe CYPs inpatient care experiences (World Health Organization, 2018: 7). An iterative process allowed for inductive analysis to develop categories of meaning (Bengtsson, 2016). 2. Drawing Inferences: Findings were applied to existing body of knowledge (Crowe et al., 2015). 3. Narrative Summary: Results were presented as a narrative summary using pseudonyms (Bengtsson, 2016). Data analysis was performed by an experienced researcher (HN) and cross-checked (HH) to increase validity and reliability. A detailed record of reasoning and reflections on the meaning and use of words by CYP was maintained to ensure trustworthiness (Bengtsson, 2016).

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Saturation was achieved when each domain was represented in the data set (Saunders et al., 2018), and overarching domains in the a priori framework were reached at the third consumer engagement focus group.

Phase 2. Face and content validity

Face validity of a new question set based on AHPEQS was assessed by four CYP Youth Advisory Group members. In an online forum, CYP reviewed, discussed, and gave verbal feedback on a revised question set with aligned direct quotes from focus groups. Three of these CYP provided written feedback on survey questions.

Using CVI, content validity of each question was assessed by six staff with expertise in consumer engagement, safety and quality, and health service management for 'relevance', 'appropriateness', 'utility', and 'question construction' against 16 quality criteria (Australian Commission on Safety and Quality in Health Care, 2017). For example, "useful to inform improvement actions". In contrast to face validity of questions by CYP, content validity informed a necessary link between experience of care and structures and process that support care as depicted in the WHO Framework (World Health Organization, 2018).

Phase 3. Construct validity analysis

Survey. As recommended by CYP, AHPEQS-Child was distributed by SMS to parents or carers of CYP aged from 12 to 17 years who had received inpatient care. Completion of AHPEQS-Child online implied passive consent. Aware of potential for answer by parent proxy, we added a screening question consistent with Hargreaves et al. (2018): "Please tell us who answered this survey": 1 (*I am the child or young person who was in hospital*), 2 (*I am parent or carer and I answered for my child*), 3 (*Both the child or young person, and my parent or carer*). Exclusion criteria were consistent with development of AHPEQS, and AHPEQS-Parent: admission for psychiatric care or substance abuse, under care of prison services, hospital admission greater than 35 days of care in one episode (Australian Commission on Safety and Quality in Health Care, 2017). AHPEQS-Child was distributed by the Health Service Business Intelligence Unit, through Research Electronic Data Capture software. Responses were monitored by health service Consumer Engagement. Data custodian approval was granted for quantitative analysis.

Sample size. Construct validity assessment used confirmatory factor analysis (CFA). Given cross-sectional data, minimum participants to variables ratio was 10:1 (Kyriazos, 2018). With 10 questions, an *a priori* sample size of 400, required a minimum of 100 surveys completed by child alone. Sample size of 100 was considered adequate if results revealed a well-determined factor with many factor loadings $\geq$.80, and communalities (R^2 in MPlus) of .50 (Tabachnick and Fidell, 2013).

Statistics. CFA assessed fit of a one-factor model of experience of care (Nelson et al., 2022; Nelson et al., 2023). Analysis began in SPSS version 25 (IBM Corp, Released 2017) for frequencies and distribution of data, and continued using MPlus statistical analysis software version 1.8.10 (Muthén and Muthén, 2017). Weighted Least Square Mean and Variance (WLSMV) estimator accounted for non-normal distribution and ordinal responses (Byrne, 2012). Factor loadings $\geq$ 0.55 or higher were considered good and were accepted if $\geq$ 0.32 (Byrne, 2012). Statistics to indicate good model fit were Root Mean Square Error of Approximation (RMSEA) <

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0.08 (90% interval including a value ≤ 0.05) (MacCallum et al., 1996). Comparative Fit Index (CFI) > 0.9; Tucker Lewis Index (TLI) > 0.9, communality (R^2>0.40) (suggesting that a factor explains 40% of total variation in a question) (Byrne, 2012). Composite reliability was calculated using Maximum Likelihood Estimator (> 0.7) (Padilla and Divers, 2015). Convergent validity of question 10 was assessed with reverse code of Net Promoter Score (Schober et al., 2018).

Correlation of NPS and the overall quality of care rating of question 10 "Overall the quality of care I received was …" used Spearman's rho to account for non-normal data distribution, $r_s \geq 0.7$ suggested a strong correlation, and $r_s \geq 0.4$ a moderate correlation (p<0.05).

FINDINGS

Phase 1. Content analysis findings.

Participants. Three FGDs were held between 5th August and 9th September 2021. A combination of online and in person attendance resulted in 182 minutes of data and 45 pages of transcript. A purposive sample included CYP aged 12 to 15 years (n=4, 2 female) and 16 to 18 years (n=8, 5 female), participants were not separated by age in FGDs. Of 12 participants, six identified as having a disability, one identified as culturally diverse, and all but one had experienced multiple inpatient admissions.

Seven participants heard of FGDs through health service consumer engagement networks, and five through health service social media. Online sound quality was poor for one online participant (William, age 12), limiting capacity to use his data. Another participant (Stella, age 18) left an online group early due to insufficient mobile device battery charge.

Fit within WHO Framework constructs. Contextually, FGD data meaningfully fit with quality statements of three a priori domains of patient experience, and demonstrated interaction with provision of care domains, and resource domains (World Health Organization, 2018). Depicted in Figure 2 with one CYP quote in each domain, these are elaborated on through qualitative findings.

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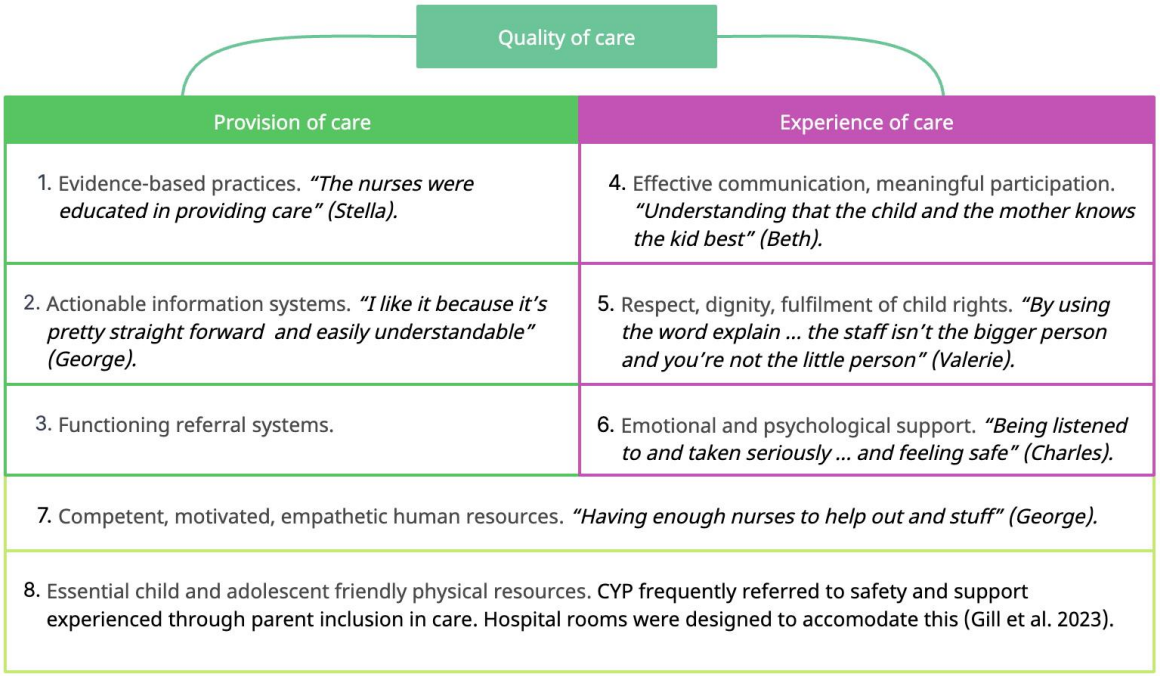


Figure 2. Qualitative findings within "WHO framework for improving the quality of paediatric care". Adapted from WHO (2018, CC BY-NC-SA 3.0 IGO.)

Provision of care.

Evidence-based practices of protection from harm, and appropriate medical care were discussed by CYP. Protection from harm overlapped across domains. Stella spoke of nurse education as most important to her experience, linked to quality statements within this domain.

"Nurses were educated in providing care" (Stella, age 18)

Actionable information systems include standardised feedback on experience of care. Children and young people valued their experience of participating in survey design. All participants preferred a CYP specific version of AHPEQS, with questions addressed in first person, demonstrated in response to AHPEQS question "I felt cared for".

"I like it because it's … straight forward and easily understandable" (George, age 15).

Changes to wording of some questions were recommended, reflecting CYPs own experience. Recommendation on AHPEQS question "I felt confident in the safety of my treatment and care" varied. Two CYP queried combining treatment and care saying they are *"very different" (Beth, age 16; Valerie age 17).* Charles (age 18) suggested that complexity was in *"confident in the safety of", rather than "treatment and care". George and William (age 12) preferred "treatment and care".*

"I think they are the same type of words and meaning" (George, age 15).

Functioning referral systems support continuity of care in a seamless and coordinated manner when CYP leave hospital, CYP did not raise this in FGDs.

Experience of care.

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Effective communication and meaningful participation. Central to this domain was CYPs request to be empowered without being overwhelmed or dismissed. Beth equated distraction techniques with loss of compassion, dignity, and emotional support, removing capacity for meaningful participation in care. Valerie asked to be included in conversations.

"Not trying to catch out kids when they feel sick or something. Not trying to distract them but understanding that the child and the mother knows the kid best" (Beth, age 16).

"Not only to speak to my parents about my health, but speak to kids about it, because it's their own health" (Valerie, age 17).

John (age 18) told of his experience of learning strategies to deal with chronic pain, and subsequent stress and experience of feeling dismissed when told *"you don't actually look in pain".* These responses cross domains at experience and provision of care levels, and into human and physical resources.

Respect, dignity, and fulfilment of child rights. Non-discrimination is "the guiding principle" of this domain (World Health Organization, 2018: 70). CYP agreed on changing AHPEQS "My views and concerns were listened to" with "I felt listened to".

"I felt listened to ... captures a bit more of the treatment of staff because there were times when I knew people were listening to me, but I didn't feel like they listened or that they cared" (Charles, age 18).

"Being listened to and taken seriously … and feeling safe" (Charles, age 18).

Feeling safe was differentiated from feeling dismissed or a perceived power imbalance:

"By using the word explain … staff isn't the bigger person and you're not the little person" (Valerie, age 17).

"*Staff carefully explained why, instead of a cold dead no" (Sara, age 18).*

This extended to agreement on adding a response option on experience of not feeling listened to for the logic flow question "my harm or distress was discussed with me by staff":

"Something about, yes we spoke about it, but I didn't feel listened to" (Charles, age 18).

"I think a lot of kids have felt that way and it would be good to have that addition in there" (Valerie, age 17).

Emotional and psychological support. Experience of care within emotional support of protective parent/carer relationships were frequently discussed by CYP, as was effective management of pain and distress. A third focus of this domain, education and play, was not raised.

Protection within parent support was reflected in discussion that to be included in their own care *"As much as I wanted is too great" (John, aged 18; Sara, age 18).*

''I really like ('included'), and I don't necessarily like 'as much as I wanted' (Janet, age 16).

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This discussion continued in response to question “I was kept informed as much as I wanted about my treatment and care”.

“Does a parent decide? … Sometimes you’re in a situation where you don’t want to know yet” (Beth, age 16).

Consensus on how to word “I received pain relief that met my needs”, was not reached by CYP, related to experience of unresolved chronic pain.

“You could have been given the pain medication, but it might not have worked” (John, age 18).

“Chronic pain, … no matter how many drugs they give you, it’s not going to completely relieve your pain” (Charles, age 18).

“It’s about helping, doing as much as they can without overstepping boundaries” (Janet, age 16).

“That’s a tricky one because I didn’t really get any pain in hospital” (George, age 15).

Human and physical resources.

Human resources, as access to sufficient trained health professionals, was named as most important by two CYP, Stella and George.

“Having enough nurses to help out and stuff” (George, age 15)

Physical environment includes amenities, medicines, play, learning, and facilities that provide for family members. CYP did not refer to play, however an importance of physical space that included family was evident in the discussions. Protection and safety were experienced when a parent was physically present to support care and decision-making.

Phase 2. Face validity and content validity findings

Face validity findings. At recommendation of CYP, minor alterations were made to some questions, and response option “I didn’t have pain” was changed to “I wasn’t in pain”. Feedback from youth included a reflection that the question-set was simple yet rigorous.

Content validity findings. Each of 16 quality criteria most frequently scored a CVI 0.83 or 1.0 representing content validity. Reviewers questioned phrasing of “treatment and care” for questions 6 and 7, asking if it represented two concepts, but not for any other question using the same phrase (questions 4, 5, 8, 9, or 10). This phrase was kept, maintaining an experience of care focus. To report on ‘care’ (without ‘treatment’) leans toward satisfaction, excluding report of experience, for example, of uncertainty around medication or information (Larson et al., 2019; Navein et al., 2022). Reviewers questioned how ethical it is to ask if CYP felt safe with treatment and care without follow up. In discussion with a child protection specialist, a logic flow link was included inviting CYP who had not felt safe to speak with their parent/carer, or Consumer Engagement (contact details were provided). Progression of changes are shown in an online supplement.

Phase 3. Construct validity findings.

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AHPEQS-Child was distributed beginning 24 October 2022. In March 2023 we observed that many parents were answering on behalf of their CYP. Of returned surveys, n=267 (17%), there was a very low response rate for surveys answered by CYP, n=34 (13%). AHPEQS-Parent survey was set to display a message on a brightly coloured screen on closing, "If your child is between 12 to 17 years of age, we will send another survey for your child to complete if they wish to do so." Following this amendment CYP most frequently answered AHPEQS-Child survey. The first n=404 (19%) of completed responses were extracted on 5 December 2023. Responses answered by parent proxy were removed, n=208 (51.9%). Data analysis reports 193 responses; CYP answered n=126 (31.4%), CYP answered with parent n=67 (16.7%). Demographic data and one data response were missing for 1 child (Table 1).

Table 1. Demographic data for assessment of construct validity.

		Frequency (percent)	Question set answered by	
			CYP alone (n=126)	CYP with parent (n=67)
Aboriginal	Yes	6 (3.1)	5	1
	No	187 (96.9)	121	66
Child's gender	Male	111 (57.5)	69	42
	Female	81 (42.0)	57	24
	Missing	1 (0.5)	0	1
Child's age	12-14 years	128 (66.3)	86	42
	15-17 years	64 (33.2)	40	24
	Missing	1 (0.5)	0	1

CFA of 10-item AHPEQS-Child model showed good fit (WLSMV estimator) (n=193): RMSEA 0.033 [90%CI 0.000, 0.064] CFI 0.998, TLI 0.997 with factor loadings all ≥ 0.476 (Table 2). Composite reliability=0.897 (Maximum Likelihood Robust estimator).

Table 2. Confirmatory factor analysis results

Question	Child alone or child with parent (n=193)		Child alone (n = 126)	
	Factor loading	R^2	Factor loading	R^2
Q1. I felt listened to	0.879	0.772	0.852	0.725
Q2. My individual needs were met	0.905	0.819	0.856	0.732
Q3. I felt cared for	0.881	0.776	0.839	0.704
Q4. I felt included in decisions about my treatment and care	0.789	0.623	0.765	0.585
Q5. I got the information I needed about my treatment …	0.805	0.648	0.777	0.604
Q6. I felt confident that all staff knew about my treatment …	0.830	0.689	0.808	0.653

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Q7. Staff helped me as much as they could when I had pain	0.476	0.226	0.501	0.251
Q8. I felt safe with the treatment and care I was given	0.839	0.704	0.782	0.612
Q9. I experienced unexpected harm or distress as a result …	0.634	0.402	0.518	0.269
Q10. Overall, the treatment and care I received was	0.917	0.840	0.905	0.820

Note. R^2 = Communality

Consistency across groups. Factor loading of Q7 "Staff helped me as much as they could when I had pain" was < 0.32 when children were admitted for medical or same-day care (Table 3). Correlation of Q10 and NPS was moderately strong, r_s 0.659 (p< 0.001).

Table 3. Model fit of 10-item AHPEQS-Child across groups.

CFA Model Q1-Q10 Child or child with parent	CFI	TLI	RMSEA [90%CI]	Factor loading ≤ 0.55	Composite reliability
AHPEQS-Child (n=193)	0.998	0.997	RMSEA 0.033 [0.000, 0.064]	0.467 (Q7-pain)	0.951
Medical (n=84)	0.997	0.997	RMSEA 0.041 [0.000, 0.092]	0.285 (Q7-pain)	0.909
Surgical (n=108)	0.992	0.989	RMSEA 0.063 [0.006, 0.100]	-	0.921
Same day (n=84)	0.997	0.997	RMSEA 0.041 [0.000, 0.092]	0.285 (Q7-pain)	0.909
Multiday (n=124)	1.000	1.000	RMSEA 0.000 [0.000, 0.053]	0.502 (Q7-pain)	0.950

Note. Composite Reliability used MLR estimator. CFI = Comparative Fit Index, TLI = Tucker Lewis Index, RMSEA = Root Mean Square Error of Approximation [90% Confidence interval].

DISCUSSION

This study aimed to understand preferences of CYP aged 12 to 18 years for reporting their own experience of inpatient health care. Qualitative methods and subsequent validation of a revised question set resulted in an age-appropriate adaptation of AHPEQS, allowing CYP to give feedback on their experience of care. Questions align with WHO 'Framework for improving the quality of healthcare' (World Health Organization, 2018). Overlap between three experience of care domains resulted in a sound one-factor model of experience of care. An interlink within eight domains of WHO Framework was evident as necessary to measure experience of care in contrast to satisfaction with care, providing actionable information to inform quality improvement (Larson et al., 2019).

Most important to CYP was to be truly listened to, heard, and acknowledged within respectful caregiving relationships. These align with UNCRC rights of participation in decisions affecting children's own life, protection from harm, and provision of care (Hammarberg, 1990). Koller et al. (2024) found that while CYPs social agency and right to participate in decision-making is recognised, a gap exists in understanding how to meet CYPs best interests and rights when considering these rights. By 12 years of age CYP generally have cognitive capacity to be included in decision-making, however their emotional state and severity of disease can affect understanding and retention of information (Grootens-Wiegers et al., 2017). Receiving too much information may cause fear and overwhelm, impeding care

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(Koller et al., 2024). In our study, CYP who had experienced chronic illness and frequent inpatient admissions discussed safety in having a trusted carer to hold knowledge and make decisions on their behalf when needed. These findings contribute insight on how to meet CYPs best interests when considering their right to protection from harm.

A right to protection from harm is balanced by communicating information to CYP who wish to grow in understanding of their own health, in a way that is meaningful and accessible (Navein et al., 2022; Davies et al., 2023), and providing enough information to invite participation, without dismissal (Koller et al., 2024). Experience of care was discussed by CYP in consideration of power dynamics. A focus on *"staff not being the bigger person and you're not the little person"* was similar to language used by children when talking of their experience of power imbalance associated with bullying (Nelson et al., 2019). Power is neutral and can be experienced with protection and support, or alternatively with harm (when misused). Davies et al. (2023) name a power dynamic between CYP and adults as a challenge to CYPs right to express their views, to be heard, and to be taken seriously in health care. This links to historical and cultural perspectives of CYPs views as less rational or less competent than adults, views that negate dignity (Navein et al., 2022). Respectful communication helps CYP to understand their condition, participate in care, and rest in emotional support of their family rather than feeling dismissed (Davies et al., 2023).

Standard six of the WHO Framework includes routine assessment and appropriate management for pain and distress (World Health Organization, 2018). Consistent with Harrison et al. (2017) and Barr et al. (2024), CYP in our study who had experienced chronic pain discussed that staff awareness of, and sensitivity to, pain management, were important to their experience of care. Although the resulting question had low communality, a focus on appropriate management of pain in the WHO Framework enhanced our understanding of study findings related to inclusion of this question. These were consistent with recommendations that although a question may have a low factor loading or low communally, it can contribute valuable information to construct that is measured (Little et al., 1999). Further demonstrated by higher factor loadings for CYP who experienced multiday or surgical admissions.

Participants in our study, aged 12 to 18 years, did not wish to add any further questions to AHPEQS-Child, however, WHO Framework standard six includes play and education. These may be important to younger children, or a different demographic of CYP (Foster et al., 2019; Barr et al., 2024). Our study was conducted in a children's hospital with a child friendly physical environment, including access to playrooms (Gill et al., 2023). This may have influenced CYP feedback. In a recent play-based study, children aged 12 years and younger reported value in play, including play as a distraction technique (Barr et al., 2024). In our study, one participant experienced distraction as trying to catch kids out, without truly listening. Distraction techniques are promoted to take young children's attention during health care (Davies et al., 2023). This raises potential for further research to study how distraction is experienced and perceived by CYP aged 12 years and over.

Strengths and limitations

There is a reported lack of precision in defining and measuring experience of care (Larson et al., 2019), and recommendation that health leaders prioritise measures of

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patient experience to assess quality of care (National Academies of Sciences and Medicine, 2018). This study contributes to filling this gap. In discussing CYPs right to be heard with due weight given to their voice, Davies et al. (2023) comment on a lack of CYPs perspectives in the literature. A strength of this study is a focus on CYPs perspective.

On distribution of AHPEQS-Child, many parents answered by proxy. Frequency of CYP answered surveys increased following a change of messaging to parents representing a strength and limitation. A strength is that our experience informs future practice in survey administration. A limitation is a low number of CYP-reported responses; however, adequate sample size and acceptable model fit were achieved. A 19% response rate is similar to other surveys addressed to parents (Hargreaves et al., 2018).

Our sample may have been limited to children motivated by provision of gift vouchers. A further limitation may be generalisability of this work beyond Australia; however, this survey was designed to an international framework. It is currently being translated into Swedish language and assessed for validity and reliability in Sweden.

IMPLICATIONS FOR PRACTICE

A clinical implication of this study is that it highlights how important it is that staff interact positively with CYP, acknowledging and valuing each, keeping them informed and where appropriate involving them in their care, while recognising a potential for overwhelm. Provision of emotional support for parents/carers should also be considered.

AHPEQS-Child provides a validated tool for CYP to report on their own experience of inpatient care, to express their voice, and be included as findings are translated into practice through a culture of active listening and respect. Guided by an international framework, AHPEQS-Child provides an indicator to inform accountable care and quality improvement action, in combination with qualitative responses. This is most attainable when interpersonal aspects of care including communication and listening, respect and protection of child rights, and emotional support exist within structural foundations that support equity.

CONCLUSIONS

This multiple-method research resulted in adaption and validation of an experience of inpatient healthcare question set (AHPEQS-Child) with and for CYP. Recognition of CYP as experts of their own experience gave credibility. Content validity included expert review of utility of AHPEQS-Child to inform improvement strategies. Construct validity, demonstrated by CFA, supports precision in measuring child-reported experience of inpatient care. A focus on differentiating experience of care from satisfaction with care, with aligned measures of child and parent report of their own experience, moves forward from measurement by parent proxy.

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Conflicts of interest

No author has expressed a conflict of interest

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Online Supplement. Progression of question set development. (Or supplementary material)

	AHPEQS (Easy English)	Proposed revision	Face validity	Content validity (Lowest rating across all criteria for each domain)	Action	Final question
Q1	My views and concerns were listened to. (Staff listened to you.)	I felt listened to. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never 6 Does not apply	*Yes* from two CYP. *Somewhat* from one CYP - preferred original.	Relevance 1.0 Appropriateness 1.0 Utility 0.83 Construction 0.83		I felt listened to. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never 6 Does not apply
Q2	My individual needs were met. (Staff made sure you got everything you needed.)	My individual needs were met. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never	*Yes* "I think the word individual is important" (all agreed)	Not assessed as consistent with previous wording.		My individual needs were met.
Q3	*Sequence guide: if Q2=Always or Mostly skip to Q4.* When a need could not be met, staff explained why (Staff told you why they could not give you what you needed)	Staff carefully explained why my needs weren't met. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never	*Yes*	Relevance 1.0 Appropriateness 1.0 Utility 0.83 Construction 0.67 for "Item uses language and syntax that are easy to understand" with a question about adding the word "carefully"	Action. CYP requested the word carefully to be added to "avoid the feeling of a cold dead no." For this reason, the wording recommended by children was maintained.	Staff carefully explained why my needs weren't met
Q4	I felt cared for. (You felt cared for)	I felt cared for. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never	*Yes*	Not assessed as consistent with previous wording.		I felt cared for.

Q5	I was involved as much as I wanted in decisions about my treatment and care. (You felt included when decisions were made about your treatment and care.)	I felt included in decisions about my treatment and care. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never	*Yes*	Relevance 0.83 Appropriateness 0.67 "Item is appropriate to all levels of health literacy" specific to definition of "treatment." Utility 1.0 Construction 0.83	Action. A definition was added for the word treatment. Treatment - like medicine, surgery, therapy	I felt included in decisions about my treatment and care Treatment - like medicine, surgery, therapy.
Q6	I was kept informed as much as I wanted about my treatment and care. (You got the information you needed about your treatment and care.)	I got the information I needed about my treatment and care. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never	*Yes*	Relevance 0.83 Appropriateness 0.83 Utility 1.0 Construction 0.67 "Item expresses concept in unambiguous terms" "Item expresses a single concept, not multiple concepts"	Action. No change to construction. "Treatment and care" were not identified as more than one concept in Q5 or Q10 or in development of AHPEQS by the ACSQHC.	I got the information I needed about my treatment and care.
Q7	As far as I could tell, the people involved in my care communicated with each other about my treatment and care. (All staff who cared for you - Worked together - Shared the right information about you.)	People involved in my care knew about my treatment and care, so I didn't have to tell them. OR I felt confident that all the staff knew about my treatment and care. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never	*Somewhat* from one CYP with preference for option 2. I felt confident that all the staff knew about my treatment and care.	Relevance 1.0 Appropriateness 0.83 Utility 1.0 Construction 0.67 "Item uses language and syntax which are easy to understand" "Item expresses concept in unambiguous terms" "Item expresses a single concept, not multiple concepts"	Action. As for Q6.	I felt confident that all the staff knew about my treatment and care
Q8	I received pain relief that met my needs. (You got pain medicine if you needed it.)	Staff helped me as much as they could when I had pain 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never 6. I didn't have pain	*Yes*, change wording of added response option to 6 "I wasn't in pain"	Relevance 0.67 "Item can apply to most consumer's health care experiences" Appropriateness 0.83 Utility 0.83 Construction 0.83	Action. Wording for added response option "I wasn't in pain"	Staff helped me as much as they could when I had pain.

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Q9	When I was in hospital, I felt confident in the safety of my treatment and care. (You felt safe when you go treatment and care)	I felt safe when I got treatment and care. OR I felt safe in the way I was cared for. 1 Always 2 Mostly 3 Sometimes 4 Rarely 5 Never	*No* from one CYP "I think I almost prefer the original wording" Action. ACSQHC "Cared for" is too open towards satisfaction.	Relevance 0.83 Appropriateness 0.33 "Item is ethically acceptable to ask without follow up" Utility 0.83 Construction 0.83	Action. 1.As most CYP said they saw treatment and care as one concept the first option was selected. 2. Discussed with child protection specialists - a logic flow pathway was added for Q9 and Q10.	I felt safe with the treatment and care I was given.
Q10	I experienced unexpected harm or distress as a result of my treatment or care. (Your treatment led to physical harm or distress that you did not expect. Harm means you get hurt. Distress means you feel upset.	I experienced unexpected harm or distress as a result of my treatment or care. 1 Yes, I was hurt. 2 Yes, I felt distress. 3 Yes, I was hurt and felt distress. 4 No, I was **not** hurt and did **not** feel distress.	*Yes*	Relevance 1.0 Appropriateness 0.5 "Item is ethically acceptable to ask without follow up" Utility 0.83 Construction 0.67 "Item expresses a single concept, not multiple concepts"	Action for Appropriateness: As for Q9. Action for construction: Wording kept as it was CYPs preferred format for the question.	I experienced unexpected harm or distress as a result of my treatment or care. Harm means you get hurt. Distress means you feel upset.
Q11	*Sequence guide: if Q10=No, skip to Q12* My harm or distress was discussed with me by staff. (Staff talked to you about the harm and distress that happened to you..	Staff talked with me about the harm or distress that happened to me. 1 Yes 2 **No** 3 **Not** sure 4 I did **not** want to talk about it **5 Yes, we spoke about it, but I didn't feel listened too.**	*Yes*, add opportunity for free text..	Relevance 1.0 Appropriateness 0.5 "Item is ethically acceptable to ask without follow up" Utility 0.83 Construction 0.67 "Item expresses a single concept, not multiple concepts"	Action for Appropriateness: As for Q9. Action for construction: Wording kept as it was CYPs preferred format for the question	Staff talked with me about the harm or distress that happened to me. 1 Yes 2 **No** 3 **Not** sure 4 I did **not** want to talk about it **5 Yes, we spoke about it, but I didn't feel listened too.**
Q12	Overall the quality of the treatment and care I received was (Overall your treatment and care in hospital was	Overall the treatment and care I received was 1 Very good 2 Good 3 OK 4 Bad 5 Very bad	*Yes*	CVI not assessed	CYPs preferred response option.	Overall the treatment and care I received was 1 Very good 2 Good 3 OK 4 Bad 5 Very bad